

Samuel Bates

Phoenix, AZ · batesambates@gmail.com · 623-694-5531

github.com/desertcache · desertcache.github.io/portfolio · linkedin.com/in/samuel-b-343205133

AI engineer at DoorDash, shipping production agent infrastructure and tooling for the 1,500-person Merchant Experience Operations organization. Path here ran through cell biology research, four years in emergency medicine, and four years at DoorDash growing from operator to engineer.

EXPERIENCE

Technical Integrations Associate (AI / Engineering) · DoorDash

Apr 2025 – present

Tempe, AZ

- ▶ Run the **AI Builder Program** for the 1,500-person Merchant Experience Operations organization. Authored the AI Builder Playbook and the Tier-Token graduation framework (\$1K, \$2.5K, \$5K monthly model-usage budgets gated by measurable improvement). Built auditing agents that review every code change against acceptable-use policy before it ships.
- ▶ Shipped four production AI applications end-to-end: **Mosaic** (an AI-assisted internal support tool serving 4,000+ support agents), **Workforce Management 2.0** (an 18-page operations dashboard covering 139 support queues across 6 outsourced vendors, with Erlang-C staffing math), **Intelligent Mapping Engine** (a full-stack platform that assigns roughly 80,000 top-tier merchants to internal account managers, replacing manual spreadsheet-based assignment workflows), and **Dev Chat AI** (a partner-facing developer assistant on developer.doordash.com). Authored roughly 57,000 net lines of code across the company's web monorepo and shipped 75+ pull requests to production in 2026 alone.
- ▶ Built and deployed **CHIP**, a generative-AI assistant for point-of-sale support specialists on the company's internal AI platform. Currently 76 monthly active users, 51 weekly, 6,846 total queries; strongest adoption of four internal AI agents. Redesigned the response architecture after early agents abandoned long outputs (1,500-word essays became 300-word, six-section structured responses with diagnostic checklists and action matrices). The new template was adopted by three additional internal agents.
- ▶ Authored business case and secured **\$91,000 in funding** for the Square V1-to-V2 migration program (approximately \$21M per month in merchant revenue at risk). Designed the operating model and the Sigma/Mode dashboards. A ten-agent vendor pod retained roughly 94% of the at-risk revenue, under budget.
- ▶ Designed and led AI-adoption workshops for roughly 1,080 leaders across four internal leadership summits. Created the *Define, Organize, Apply, Refine* framework that seeded five to seven cross-organization AI partnerships.
- ▶ Built and operate a **six-agent local production AI fleet** (Atlas, Argus, Hermes, Athena, Oracle, Mnemosyne) with autonomous dispatch, scheduled context loops, and a self-correcting memory system, running as macOS background services.
- ▶ Two Merchant Services Excellence Awards for AI workshop delivery and migration training impact.

Disaster Prevention Specialist · DoorDash

Nov 2023 – Apr 2025

Tempe, AZ

One of four specialists handling executive-level escalations and high-severity cases (50+/day) across consumer, driver, and merchant lines. Built SQL tools, Salesforce reports, and prevention playbooks; fraud investigation and platform-integrity work.

Tempe, AZ

Designed standard operating procedures that extended automated point-of-sale provider coverage from 1 to 23 systems and improved specialist-to-provider ratio from 1:1 to 1:7. Tested and integrated 9,100+ Otter and Toast merchant stores; total integrated stores scaled to 39,114.

EMT · Behavioral Health Specialist Team Lead · Phoenix Fire Department

Mar 2020 — Sept 2022

Phoenix, AZ

Led a 12-person crisis response team coordinating with fire, medical, and law enforcement on emergency dispatches. On-site crisis intervention; mentored and trained new team members in advanced crisis management.

TECHNICAL SKILLS

Languages	Python · TypeScript / JavaScript · SQL · C++ · Bash
AI / ML	Frontier large-language-models in daily production across all three major providers: Anthropic Claude (Opus 4.7, Sonnet 4.6, Haiku 4.5), OpenAI ChatGPT (GPT-5.5), Google Gemini (3.1 Pro) · Anthropic Agent SDK with vision, computer use, and the Model Context Protocol · Bayesian experiment design · local LLM inference (Ollama, Apple MLX)
Frontend	React 18/19 · Redux · Zustand · React Query · Three.js · Recharts · Tailwind · shadcn/ui
Backend	Node.js · Fastify · Express · GraphQL (Apollo) · REST APIs · Prometheus monitoring
Data	Snowflake (advanced SQL) · pandas · Streamlit · Sigma · Mode
Desktop	Electron · py2app · macOS background services · ffmpeg
DevOps	GitHub Actions · BuildKite · Docker · monorepo build systems

EDUCATION

B.S. Cell & Molecular Biology, Minor in Chemistry · Western New Mexico University · 2019

Undergraduate research: manganese-II accumulation and efflux-gene sequencing/cloning in Deinococcus radiodurans — molecular basis of extremophile radiation tolerance.

NREMT-B Emergency Medical Technician · Estrella Mountain Community College